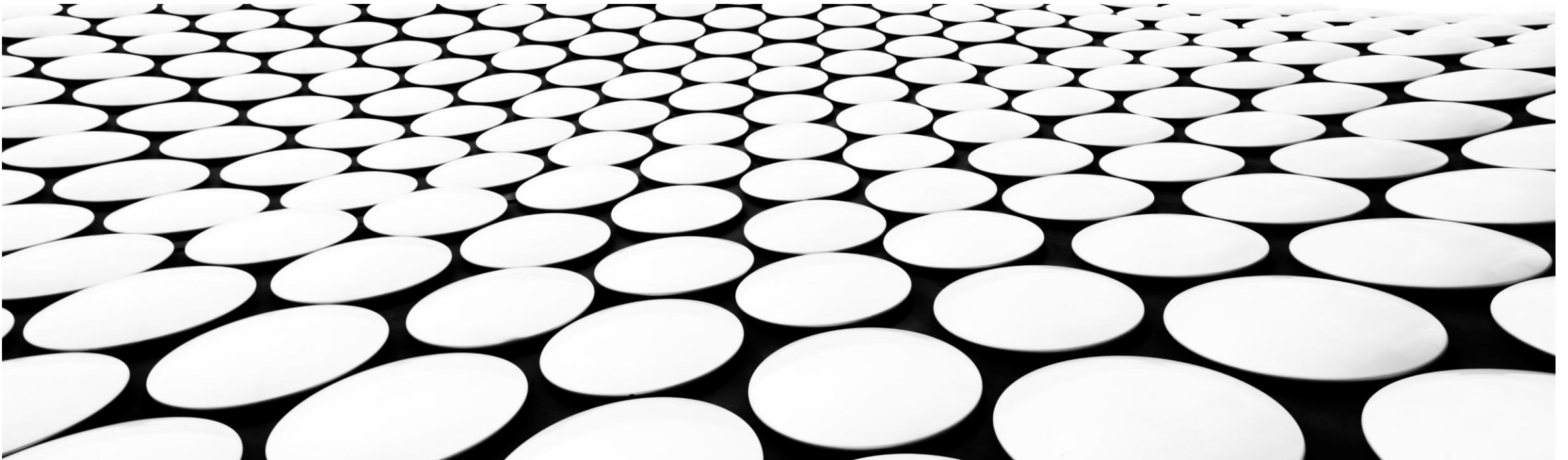


IOT DESIGN AND PROTOTYPING (420-N55-LA)





INTRODUCTION TO THE COURSE



COURSE INFORMATION

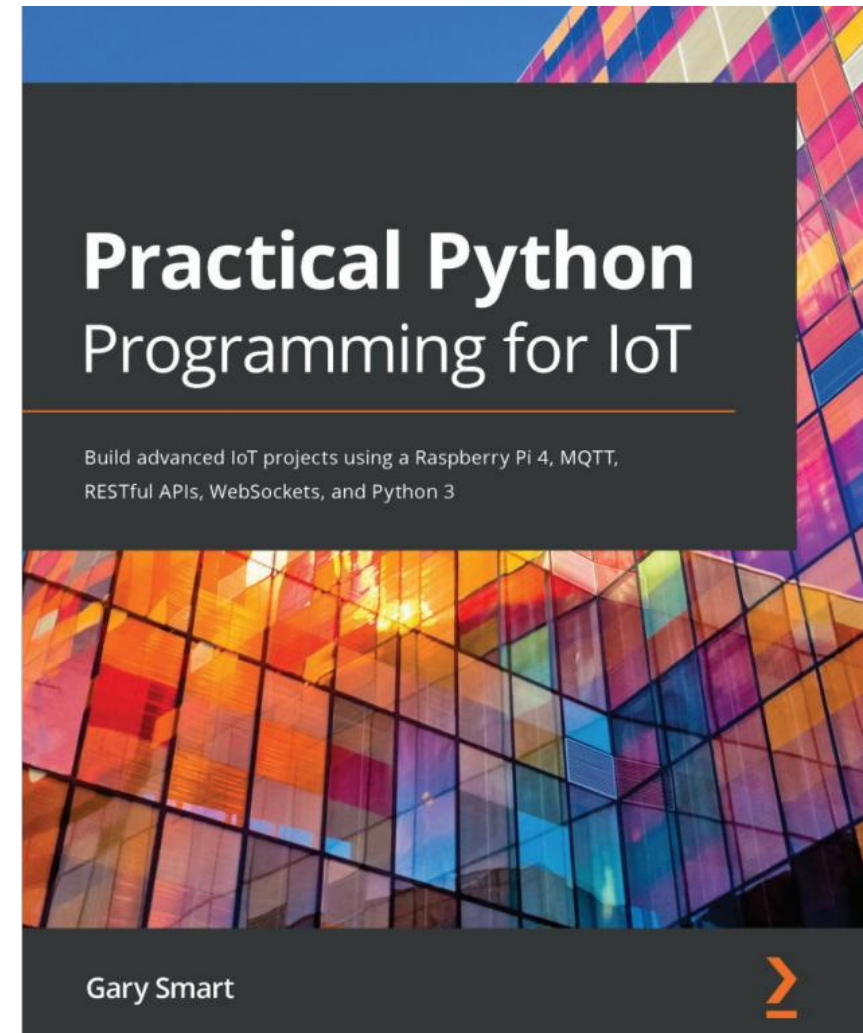
Course title	IoT: Design and Prototyping of Connected Devices		
Course number	420-N55-LA		
Credits:	1.33		
Hours:	45		
Weighting:	1-2-1		
Modality:	Classroom and laboratory / In-Person		
Prerequisites:	420-N35-LA Introduction to IoT (Internet of Things) and Embedded Systems	Co-requisites:	

OVERVIEW

- This course explores the design and creation of interconnected digital ecosystems. Starting with a solid understanding in n-tier application design, students will be introduced to Raspberry-Pi, the adaptable Raspbian OS, Docker, MySQL, and MongoDB. Beginning with essential Python basics, the curriculum advances to detailed studies of Python paradigms specifically crafted for IoT.
- Participants will navigate the complexities of the Raspberry Pi platform, unlocking its potential with Python-driven GPIO operations.
- They will also grasp the essence of web services, utilizing Python as both a client and a service, and master RESTful web service connections as well as Spring-based implementations.
- This journey enhances students' understanding of databases, emphasizing the integration of MySQL with Python for optimal data storage. As students stride towards a secure digital future, they'll be equipped to fortify their IoT projects against potential threats.
- The course culminates in hands-on project presentations, celebrating their foray into the IoT realm.

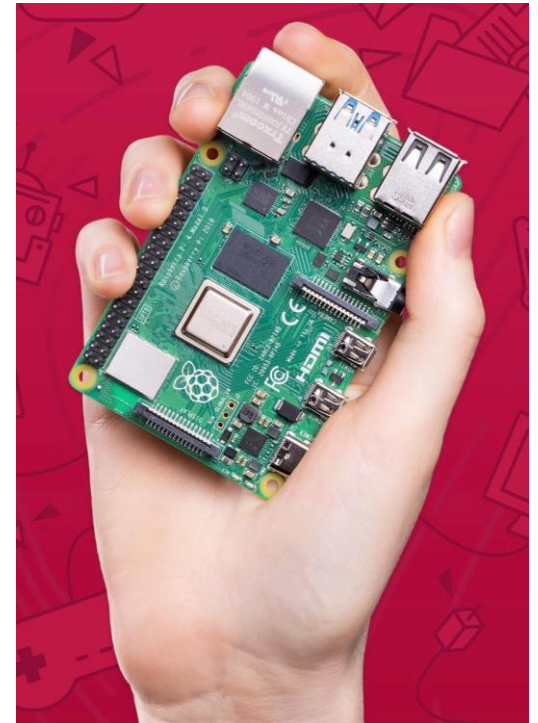
COURSE MATERIAL

- Textbook
- Practical Python Programming for IOT by Gary Smart. Packt Publishing 2020. ISBN 978-1-83898-246-1



REQUIRED HARDWARE

1. PC/LAPTOP
 - You need this for basic development much like your other courses.
 - Mac or PC, whatever.
2. Raspberry Pi
 - Case
 - Memory Card
 - Power Supply
3. Electronics
 - Temperature/Humidity Sensor
 - LED's, Buttons, Resistors
 - OLED Display
4. Monitor and Keyboard –for home use.



RASPBERRY PI 5

Find deals here: <https://rpilocator.com/>

Raspberry Pi SKU: 734053 Mfr Part#: SC1642

Raspberry Pi 5

Broadcom BCM2712 Quad-Core Cortex-A76; 2GB LPDDR4X RAM

★★★★★ 4.9 (9)



Key Features

- Broadcom BCM2712 Quad-Core Cortex-A76
- 2GB LPDDR4X RAM

Protect and Support ⓘ

Free 60 Day Tech Support

Add Ons

- ☐ Micro Center 32GB microSDHC Class 10 / U1 Flash Me... \$36.99
was \$50.99
- ☐ Sandisk Ultra PLUS 32GB microSDHC Class 10/ V10 FL... \$11.49
was \$12.99
- ☐ Schmartboard Inc. 9" Male to Female Jumper Wires wit... \$7.99

MAY BE RETURNED WITHIN 30 DAYS OF PURCHASE* [Learn More](#)

1 at \$49.99 each
2+ at \$49.99 each

18 MINUTE PICKUP
Not Available

MAP YOUR TRIP
Add to List
ANYTIME

SHIPPING
Ship This Item
FASTEST DELIVERY
AUG. 22

Usually ships in 5-7 business days.
 Shippable Items Text to Me



Raspberry Pi 5 ABS Case Black Silver Plastic Shell Optional PWM Cooling Fan Active Cooler Support 3.5 Inch Screen for RPi 5 Pi5

★★★★★ 4.7 31 Reviews | 331 sold

BACK TO SCHOOL • Welcome deal

Starts: Aug 18, 03:00 EST

C\$3.28 Sale is coming soon
~~C\$10.34~~

Tax excluded, add at checkout if applicable; Extra 2% off with coins

Color: Black-ActiveCooler



https://www.aliexpress.com/item/1005007522341772.html?spm=a2g0o.productlist.main.53.526e5057d1T173&algo_pvid=9bdfd7f2-208d-48e0-a435-14332f15af19&algo_exp_id=9bdfd7f2-208d-48e0-a435-14332f15af19-48&pdp_ext_f=%7B%22order%22%3A%22331%22%2C%22eval%22%3A%221%22%7D&pdp_npi=6%40dis%21CAD%215.44%211.40%21%21%213.85%210.99%21%402101e9ec17553879336035352e14ab%2112000041129421514%21sea%21CA%210%21ABX%211%210%21m03_new_user%3A-29895%3Bn_tag%3A-29910%3BpisId%3A5000000174127056&curPageLogUid=qVaxhLTPESjL&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005007522341772%7Cp_origin_prod%3A

<https://www.microcenter.com/product/683269/raspberry-pi-5>

RASPBERRY PI 5

Electronics › Computers & Accessories › Computer Components



[Click to see full view](#)

CanaKit Raspberry Pi 5 Starter Kit PRO - Turbine Black (128GB Edition) (8GB RAM)

Brand: CanaKit

4.6 ★★★★★ (1,384)

Amazon's Choice for "raspberry pi 5 amazon"

50+ bought in past month

\$349⁹⁹

Size: 8GB RAM

Brand	CanaKit
Model name	Raspberry Pi 5
RAM memory installed size	8 GB
Memory storage capacity	128 GB
CPU model	MediaTek MT8125
CPU speed	2.4 GHz

▼ [See more](#)

IOT KITS

Raspberry Pi 4 Kit 8G 4G 2G 1 GB RAM + Case + Fan + Heat Sinks + Power Adapter Optional 32 64 128GB TF Card for RPI 4

★★★★★ 4.7 26 Reviews | 160 sold

Welcome deal

C\$102.11 New shoppers save C\$22.60
C\$124.71

Tax excluded, add at checkout if applicable

Bundle: Bundle 5



Color: 2GB RAM

1GB RAM

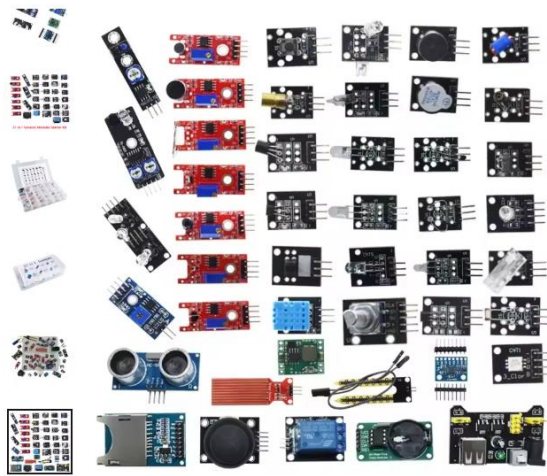
2GB RAM

4GB RAM

8GB RAM

https://www.aliexpress.com/item/1005008228051693.html?spm=a2g0o.productlist.main.48.3d7dYm9kYm9kB0&algo_pvid=84a916f0-1550-42ed-b878-f3475f806227&algo_exp_id=84a916f0-1550-42ed-b878-f3475f806227-45&pdp_ext_f=%7B%22order%22%3A%22160%22%2C%22eval%22%3A%221%22%7D&pdp_npi=6%40dis%21CAD%2189.18%2166.58%21%21%2163.12%2147.12%21%402103245417553826329864492e7f85%2112000044309916547%21sea%21CA%210%21ABX%211%210%21m03_new_user%3A-29895%3Bn_tag%3A-29910%3BpisId%3A5000000174127041&curPageLogUid=TeGxuHP0zQKQ&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005008228051693%7C_p_origin_p rod%3A

IOT SENSORS



45 in 1 Sensors Modules Starter Kit for arduino, better than 37in1 sensor kit
37 in 1 Sensor Kit whit box

★★★★☆ 4.4 118 Reviews | 1,000+ sold

BACK TO SCHOOL • Welcome deal

Starts: Aug 18, 03:00 EST

C\$14.82 Sale is coming soon
~~C\$23.30~~

Tax excluded, add at checkout if applicable ; Extra 5% off with coins

Color: 45 in 1 kit



https://www.aliexpress.com/item/4000267655567.html?spm=a2g0o.order_detail.order_detail_item.2.30a2f19cJ01X2Y

OTHER ACCESSORIES



Raspberry Pi 40 Pin GPIO Extension Board + MB-102 Breadboard PCB
Bread Board + GPIO Cable Adapter for Orange Pi for Arduino UNO

★★★★★ 5.0 6 Reviews | 68 sold

BACK TO SCHOOL • Welcome deal

Starts: Aug 18, 03:00 EST

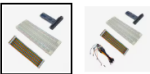
C\$5.18

Sale is coming soon

C\$9.42

Tax excluded, add at checkout if applicable ; Extra 2% off with coins

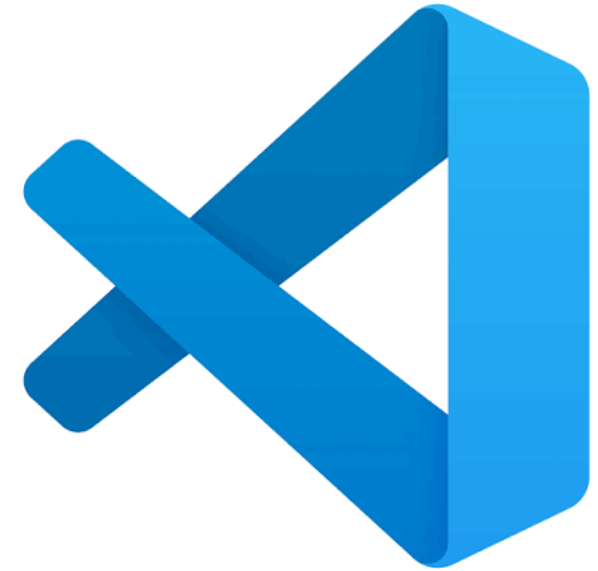
Color: Kit 1



https://www.aliexpress.com/item/32596465282.html?spm=a2g0o.detail.pcDetailTopMoreOtherSeller.12.1540tVAAtVAAFx&gps-id=pcDetailTopMoreOtherSeller&scm=1007.40050.354490.0&scm_id=1007.40050.354490.0&scm-url=1007.40050.354490.0&pvid=51dd73a6-a59f-4ca8-9433-d714dfa3e4ce&t=gps-id:pcDetailTopMoreOtherSeller,scm-url:1007.40050.354490.0,pvid:51dd73a6-a59f-4ca8-9433-d714dfa3e4ce,tpp_buckets:668%232846%238111%231996&pdp_ext_f=%7B%22order%22%3A%2268%22%2C%22eval%22%3A%221%22%2C%22sceneId%22%3A%2230050%22%7D&pdp_npi=6%40dis%21CAD%219.42%215.18%21%21%216.67%213.67%21%402103205117553890001803750ed04a%2166120558428%21rec%21CA%21%21ABXZ%211%210%21m03_new_user%3A-29895%3Bn_tag%3A-29910%3BpisId%3A5000000174127034&utparam-url=scene%3ApcDetailTopMoreOtherSeller%7Cquery_from%3A%7Cx_object_id%3A32596465282%7C_p_origin_prod%3A

MEDIAGRAPHY/SOFTWARE

- [Visual Studio Code](#) (commonly known as VS Code) is a free, lightweight, and highly customizable source code editor developed by Microsoft.
- It is currently the most popular development environment in the world, supporting Windows, macOS, Linux, and web browsers. [[1](#), [2](#), [3](#), [4](#), [5](#)]



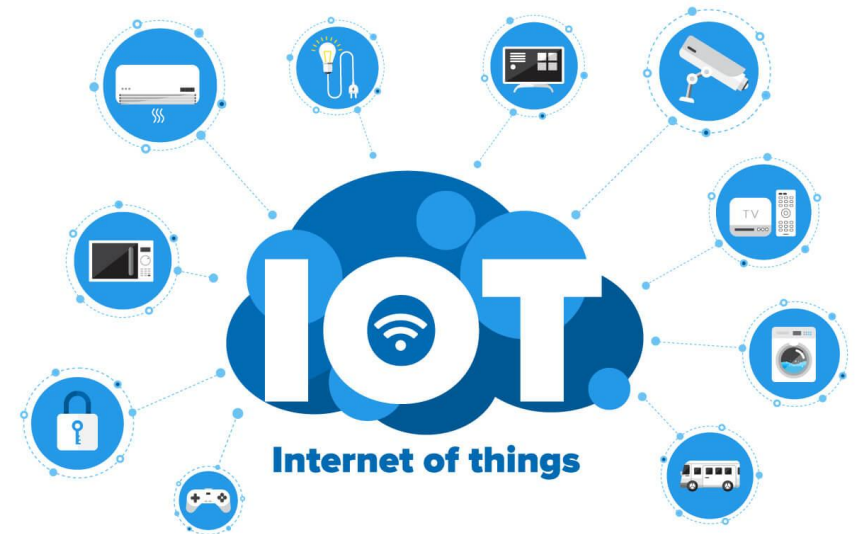


INTERNET OF THINGS



WHAT IS INTERNET-OF-THINGS (IOT)

- The Internet of Things (IoT) refers to the network of interconnected physical devices, vehicles, buildings, and other objects that are embedded with sensors, software, and other technologies, allowing them to collect and exchange data over the internet.
- The concept behind IoT is to enable these devices to communicate and interact with each other and with central systems, often without requiring human intervention.



TOP INTERNET-OF-THINGS (IOT) EXAMPLES TO KNOW

- Connected cars.
- Smart appliances.
- Connected security systems.
- Smart agriculture equipment.
- Connected retail.
- Connected healthcare monitors.
- Connected manufacturing equipment.
- Connected cities.
- Watch:

<https://www.youtube.com/watch?v=7yYb2224yH0>



IOT PROJECTS USING ARDUINO & RASPBERRY PI

- IoT Pollution monitoring
- IoT Color based sorting (garbage sorting)
- IoT Weather Station monitoring
- IoT ICU patient monitoring
- IoT smart dust bin
- IoT car parking monitoring and automation

Watch:

<https://www.youtube.com/watch?v=ZOmlaimMaJc>

<https://www.youtube.com/watch?v=WINVetH4ieg>

https://www.youtube.com/watch?v=_AlcRoqS65E&t=10s